

10th Mar 2026 | DA6360

⇒ General Quadratic Funcⁿ

$$f(x) = \frac{1}{2} x^T H x + g^T x + c$$

$H \in \mathbb{R}^{d \times d}$ is a sym. +ve definite matrix

$$\nabla_x f(x) = Hx + g$$

$$\text{Pick } H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad g = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_{t+1}(1) \\ x_{t+1}(2) \end{bmatrix} = \begin{bmatrix} x_t(1) \\ x_t(2) \end{bmatrix} - \alpha \left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_t(1) \\ x_t(2) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$$

$$x_{t+1} = \begin{bmatrix} 1-2\alpha & -\alpha \\ -\alpha & 1-2\alpha \end{bmatrix} x_t - \alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_{t+1} = x_t - \alpha (Hx_* + g)$$

$$x_{t+1} - x_* = x_t - x_* - \alpha (H(x_t - x_* + x_*) + g)$$

$$e_{t+1} = (I - \alpha H) e_t$$

$$e_t = (I - \alpha H)^t e_0$$

Spectral Theorem $H = U \Lambda U^T$

$$\begin{aligned} e_{t+1} &= (I - \alpha U \Lambda U^T) e_t \\ &= (UU^T - \alpha U \Lambda U^T)^t e_t \end{aligned}$$

$$c_{t+1} = U (I - \alpha \Lambda) U^T c_t$$

$$c_{t+1} = U (I - \alpha \Lambda) U^T c_t$$

$$U^T c_{t+1} = U^T \cdot U (I - \alpha \Lambda) U^T c_t$$

$$U^T c_{t+1} = (I - \alpha \Lambda) U^T c_t$$

define,

$$z_t = U^T c_t$$

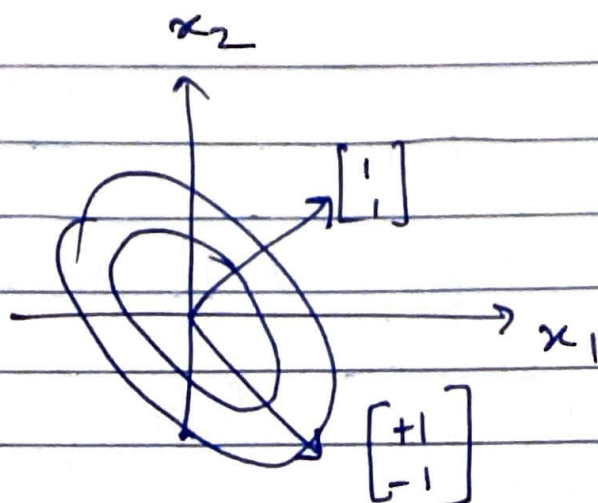
↪ error measured in orthonormal basis U .

$$z_{t+1} = (I - \alpha \Lambda) z_t$$

$$z_t = (I - \alpha \Lambda) z_t$$

now how can I express this diagonometrically -

say $g = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$



$\Rightarrow$ SGD (Stochastic GD)
(workhorse of modern ML/AI)

- eg. logistic regression, linear reg.
 $(x_i, y_i)_{i=1}^n \in \mathbb{R}^d \times \{-1, +1\}$, $(x_i, y_i)_{i=1}^n \in \mathbb{R}^d \times \mathbb{R}$

- dataset has i to n pts. & each pt. has its own loss funcⁿ $f_i(\theta)$.

- then the overall loss $f(\theta) = \frac{1}{n} \sum_{i=1}^n f_i(\theta)$

- linear model is

$$\hat{y}(x) = \theta^T x$$

$$f_i(\theta) = \frac{1}{2} (\theta^T x_i - y_i)^2$$

$$f(\theta) = \frac{1}{2n} \sum_{i=1}^n (\theta^T x_i - y_i)^2$$

$$\nabla_{\theta} f(\theta) = \frac{1}{2n} \sum_{i=1}^n 2(\theta^T x_i - y_i) x_i$$

- take an eg. of imagenet

$$x \in \mathbb{R}^d$$

$$d = 300 \times 300 \times 3 \\ \approx 30,000$$

$$n = 10^6$$

d & n are in the order of 10^6 , so we do not want to sum large no. of vectors
sample a subset $I \subseteq \{1, \dots, n\}$ with high dims.
batch

$$\nabla f(\theta_t) \approx \hat{\nabla} f(x_t) = \frac{1}{|I|} \sum_{i \in I} \nabla f_i(\theta_t)$$

I elements are chosen at random.

$$|I| < 100 \quad \text{or} \quad |I| = 128 \quad \text{or} \quad 32$$

- stochastic gradient

$$E[\hat{\nabla} f(\theta_t)] = \nabla f(\theta_t) \Rightarrow \text{that's why c/d stochastic}$$

$$\theta_{t+1} = \theta_t - \alpha \nabla f(\theta_t) \quad (\text{GD})$$

$$\theta_{t+1} = \theta_t - \alpha \hat{\nabla} f(\theta_t) \quad (\text{SGD})$$

- analysis of $\theta_{t+1} = \theta_t - \alpha \hat{\nabla} f(\theta_t)$

$$\theta_{t+1} = \theta_t - \alpha (\hat{\nabla} f(\theta_t) + \nabla f(\theta_t) - \nabla f(\theta_t))$$

$$= \theta_t - \alpha \left(\nabla f(\theta_t) + \underbrace{\hat{\nabla} f(\theta_t) - \nabla f(\theta_t)}_{N_t} \right)$$

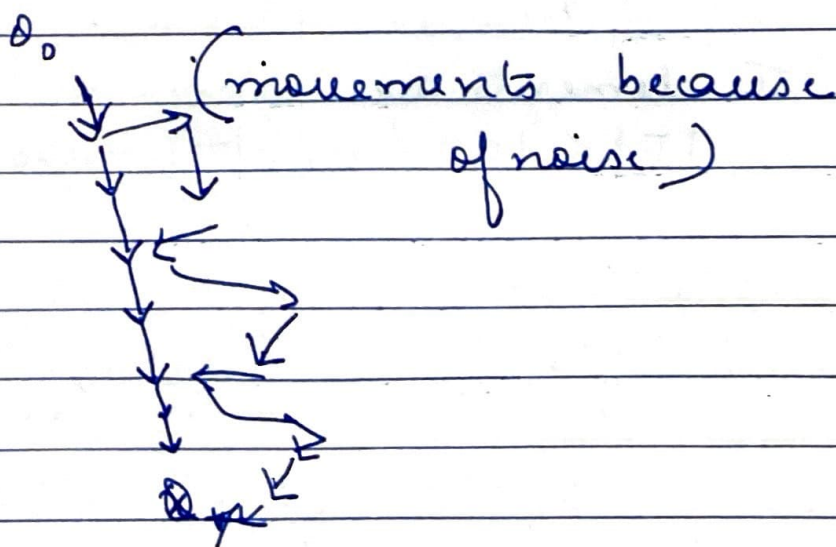
N_t = is a 0 noise mean term
 $E[N_t] = 0$

$$\theta_{t+1} = \theta_t - \alpha \left(\nabla f(\theta_t) - \alpha N_t \right)$$

If N_t is 0, then
 SGD is GD.

For GD

For SGD



$$E[\theta_{t+1}^{SGD}] = E[\theta_t^{SGD}] - \alpha E[\hat{\nabla} f(\theta_t^{SGD})]$$

← $\theta_{t+1}^{SGD} = \theta_t^{SGD} - \alpha \left(\hat{\nabla} f(\theta_t^{SGD}) \right)$

expected trajectory

$\sigma^2 \rightarrow$ variance (N_t)

$$\text{Total noise} = \alpha^2 \sigma^2 + \alpha^2 \sigma^2 + \dots + \alpha^2 \sigma^2.$$

- For SGD to converge, we need diminishing step-size (LR).

$$\begin{aligned} \theta_{t+1} &= \theta_t - \alpha_t \hat{\nabla} f(\theta_t) + N_t \\ \alpha_t &\rightarrow 0 \\ \sum \alpha_t &\rightarrow \infty \end{aligned}$$

Since we reduce α , then the radii of my noise will reduce, the bounding region of my noise will also reduce.

- Stochastic Approximation Conditions

$$\begin{aligned} \rightarrow \alpha_t &\rightarrow 0 & \sum \frac{1}{t} &= \infty & \text{generally LR is constant} \\ \rightarrow \sum \alpha_t &\rightarrow \infty & & & \text{\& scheduled} \\ \rightarrow \sum \alpha_t^2 &< \infty & \sum \frac{1}{t^2} &< \infty & \text{down-scaling its value} \end{aligned}$$